

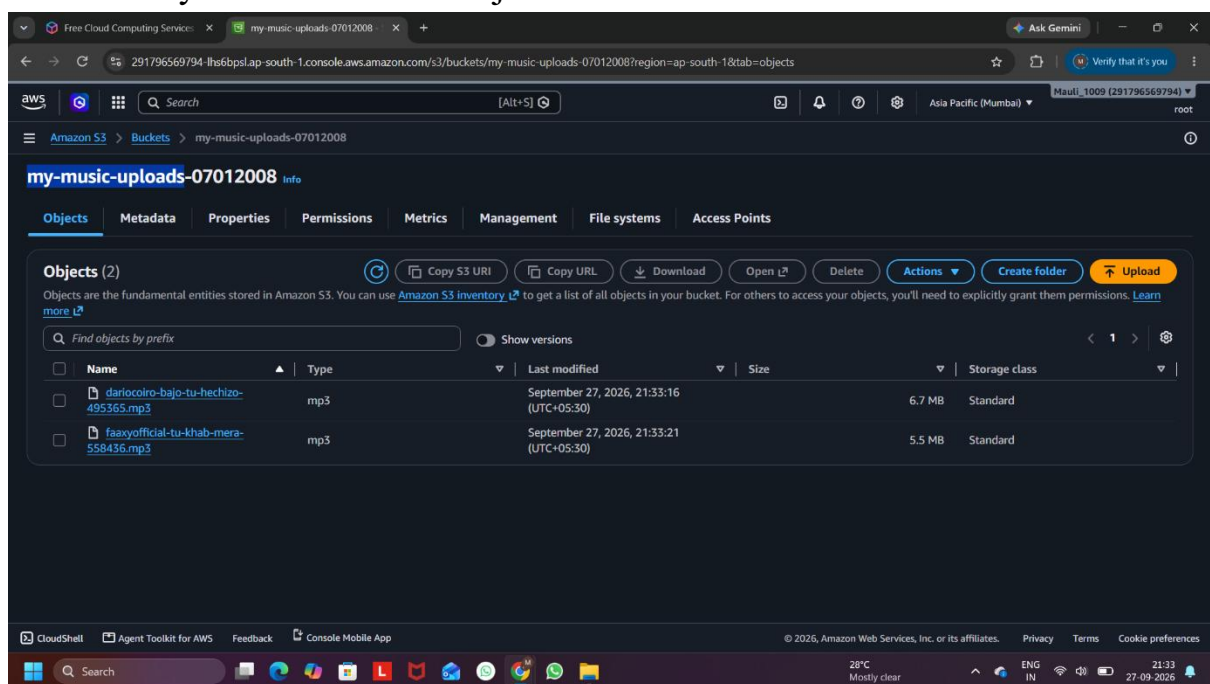
Cloud-Storage-Assignment-2

Q1. Create a new storage bucket in your chosen cloud provider (AWS S3, Google Cloud Storage, or Azure Blob Storage) named 'my-music-uploads' and upload any 2 MP3 files to it.

Ans:

Steps:

1. Open Amazon S3 and create a bucket named my-music-uploads.
2. Open the bucket and upload any two MP3 files.
3. Verify both files in the Objects tab.



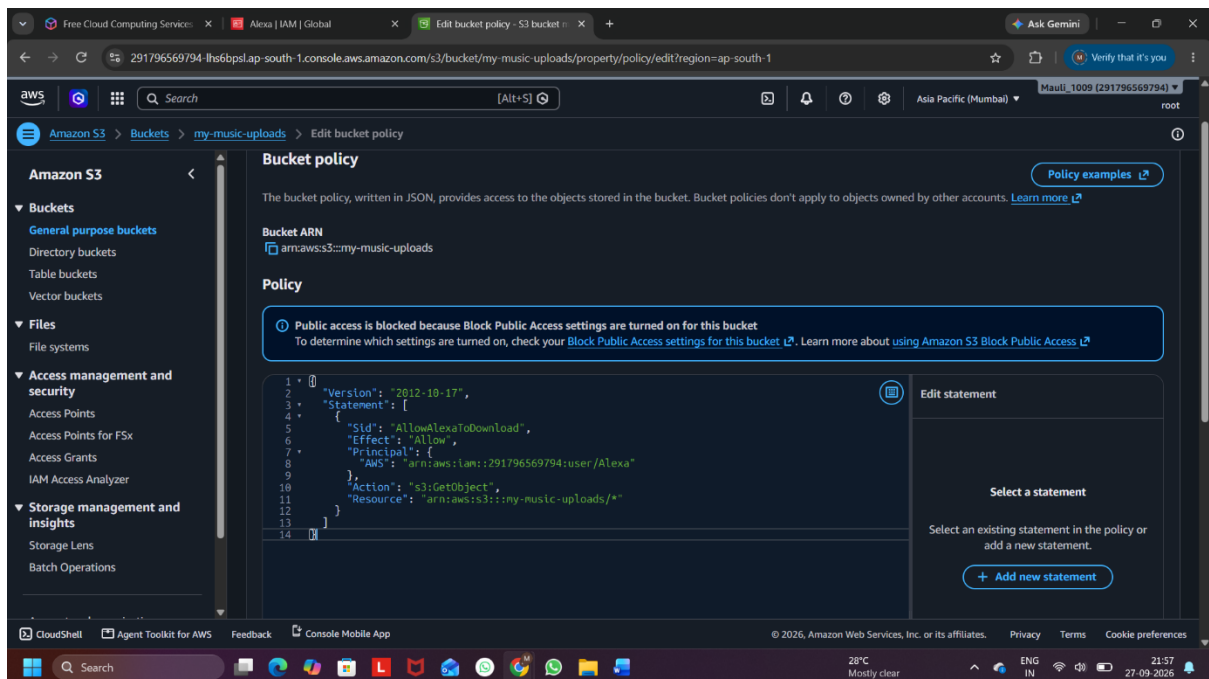
Screenshot 1: Successfully Created S3 Bucket and Uploaded Two MP3 Files.

Q2. Apply a bucket policy to 'my-music-uploads' so only users with a specific email address (your own or a friend's) can download files from the bucket. Hint: Use the bucket's access management settings to add an IAM user or assign a permission for a specific email.

Ans:

1. Open S3 bucket and navigate to the Permissions tab.
2. Add a bucket policy allowing the specific IAM user to download objects.

3. Save and verify the bucket policy.



Screenshot 1: Successfully Applied Bucket Policy for Specific IAM User

Result:

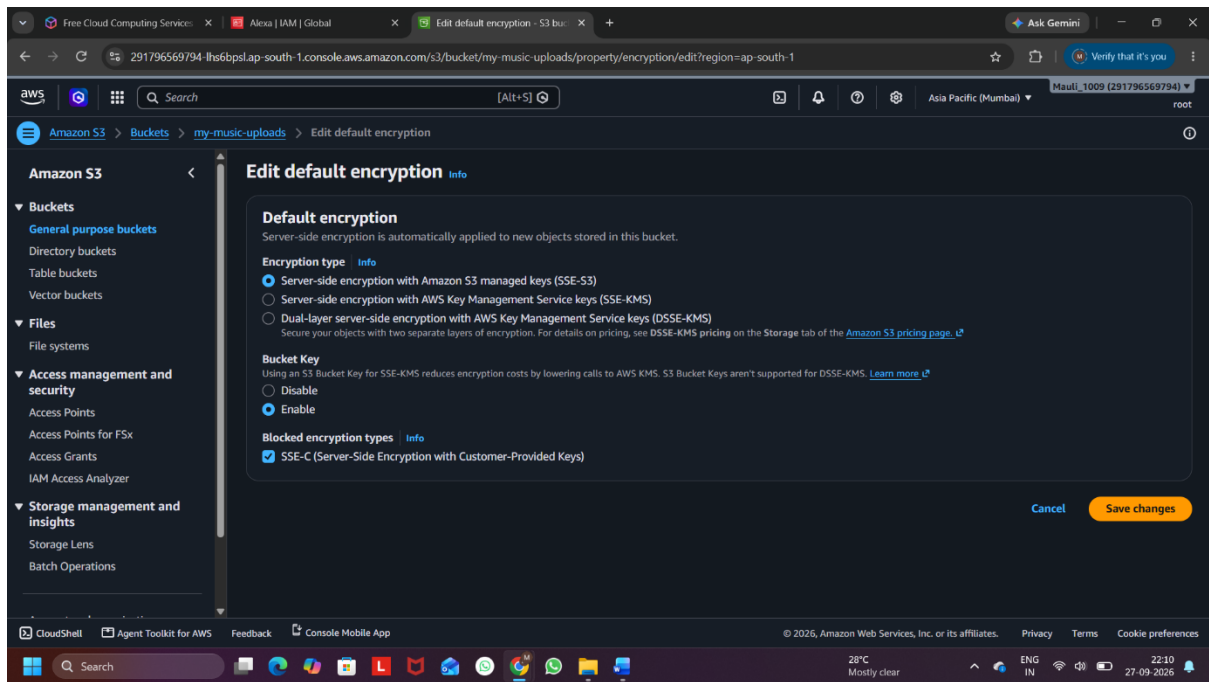
The bucket policy was applied to allow the specified IAM user to download files.

Q3.Enable server-side encryption for all objects uploaded to 'my-music-uploads', ensuring that future uploads are encrypted automatically by the cloud provider.

Ans:

Steps:

1. Open the S3 bucket and go to the Properties tab.
2. Enable default encryption using SSE-S3.
3. Save changes to automatically encrypt future uploads.



Screenshot 1: Enabled Default Server-Side Encryption (SSE-S3).

Result:

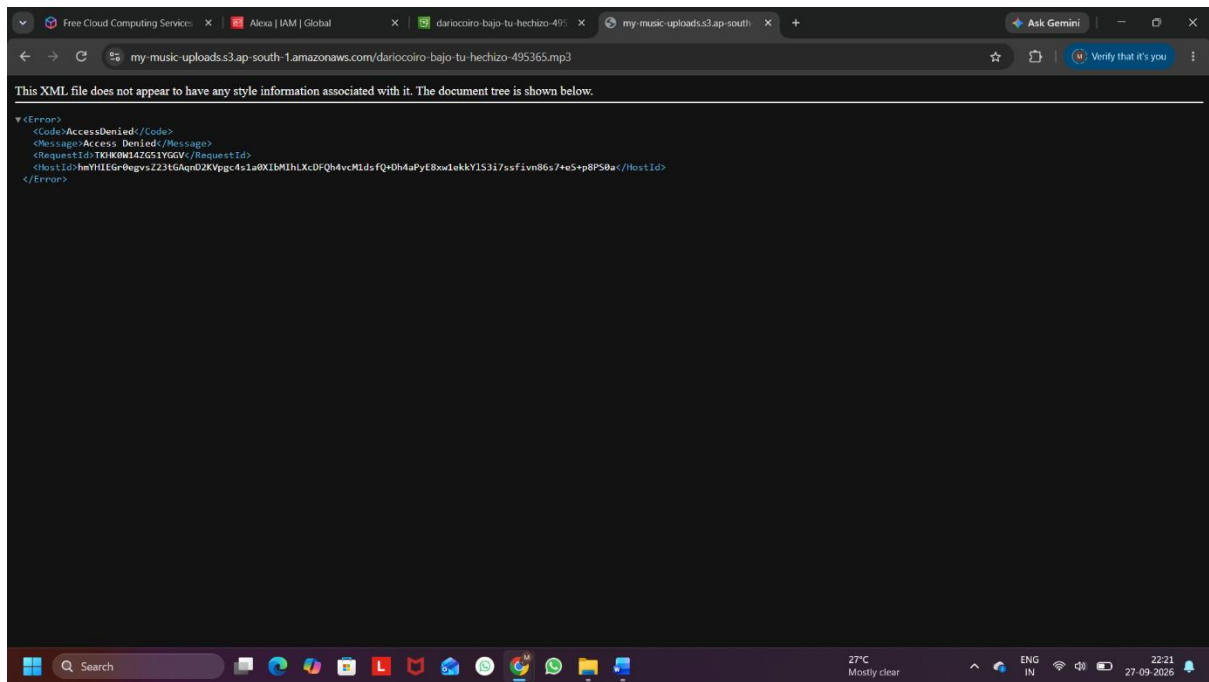
All future files uploaded to the bucket will be automatically encrypted by AWS.

Q4. Test your bucket's security by trying to access an uploaded MP3 file from a browser where you are not logged in — take a screenshot of the error or access denied message.

Ans:

Steps:

1. Copied the MP3 object URL from the S3 bucket.
2. Opened the URL in a browser without logging in.
3. Verified that the “**AccessDenied**” error was displayed.



Screenshot 1: Testing S3 Bucket Security – Access Denied for Unauthorized Users

Result:

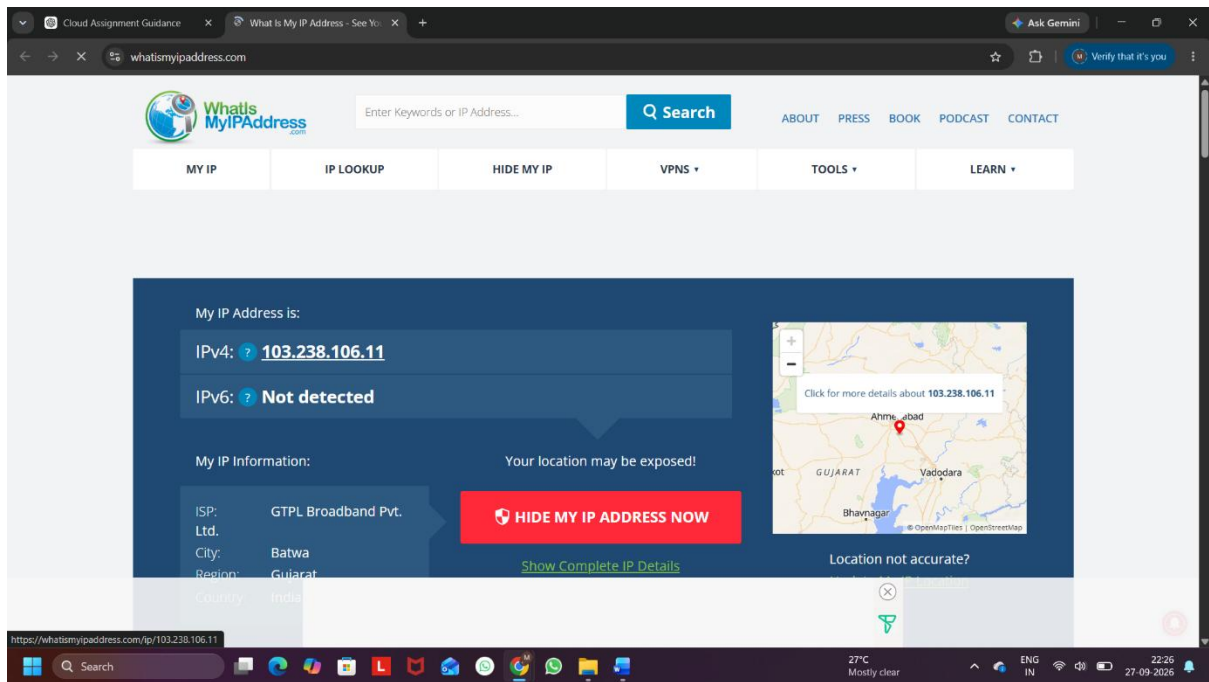
Unauthorized access to the MP3 file was denied, confirming that the file is not publicly accessible.

Q5. Use an AI tool like ChatGPT or Copilot to generate a sample bucket policy that restricts access to your bucket by IP address, then modify and apply it to 'my-music-uploads'. Hint: Copy the AI-generated policy, review it for your cloud provider's syntax, and adjust the IP to your college's or your own network.

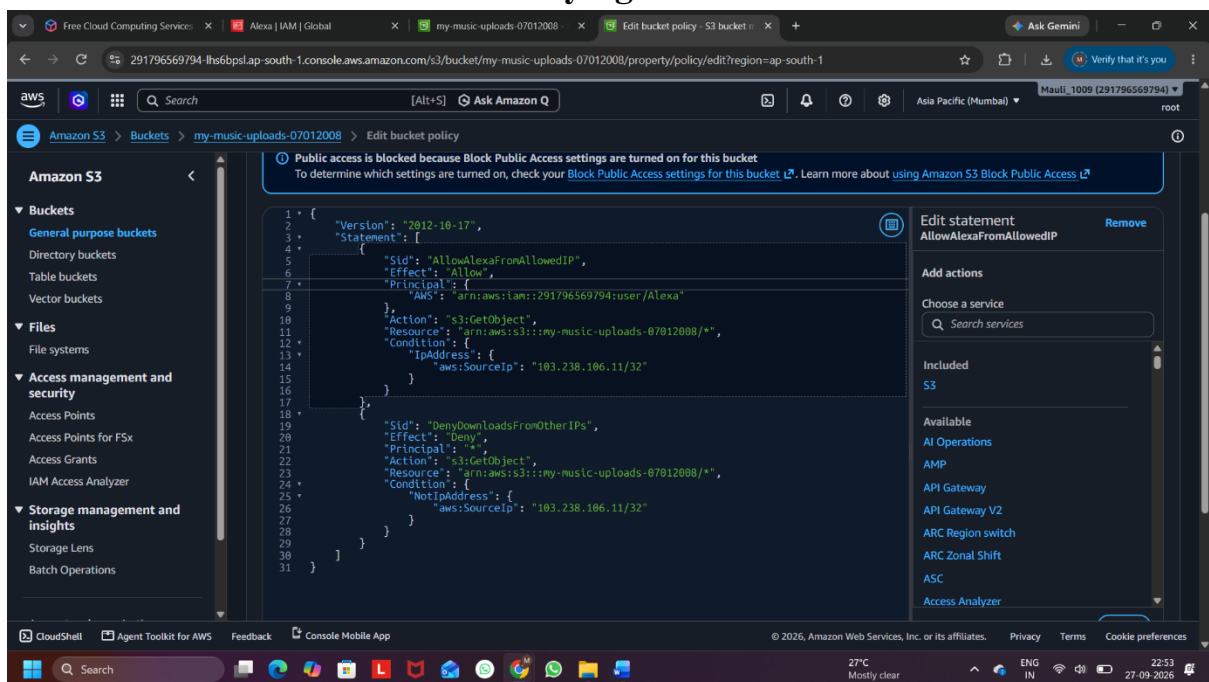
Ans:

Steps:

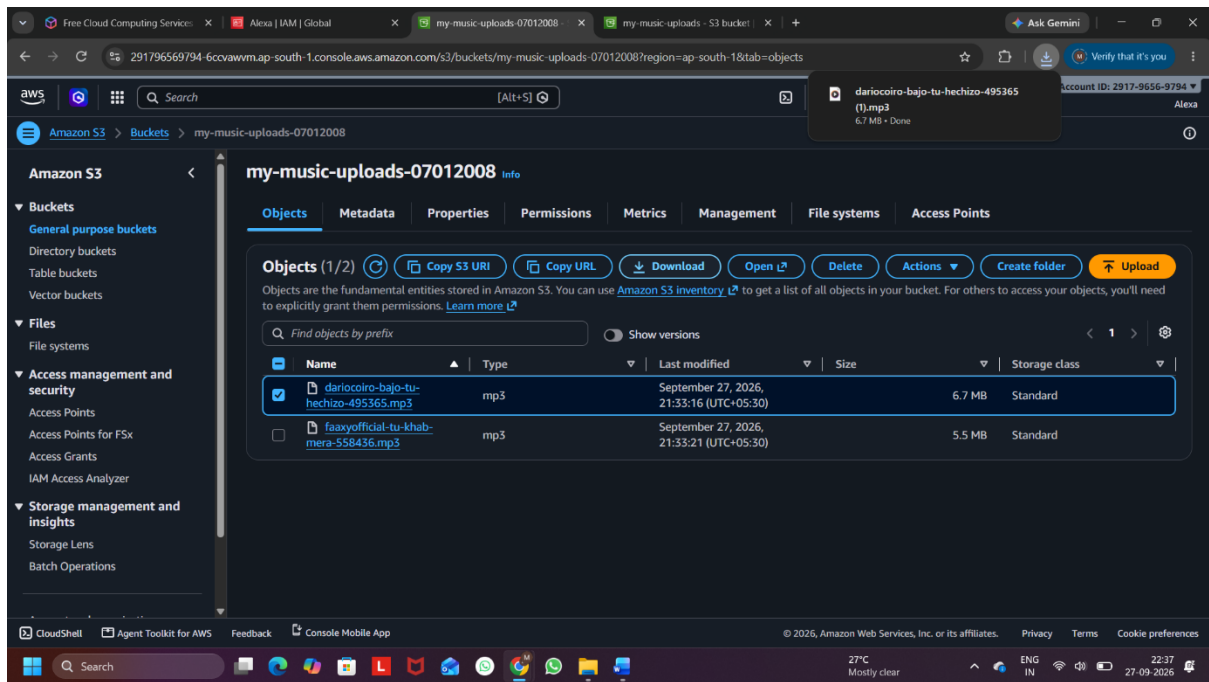
1. Used ChatGPT to generate an S3 bucket policy and modified it with the authorized IP address.
2. Applied the policy to the S3 bucket to allow access only from the specified IP address.
3. Tested access by downloading an MP3 from the authorized IP and accessing it from another IP address.



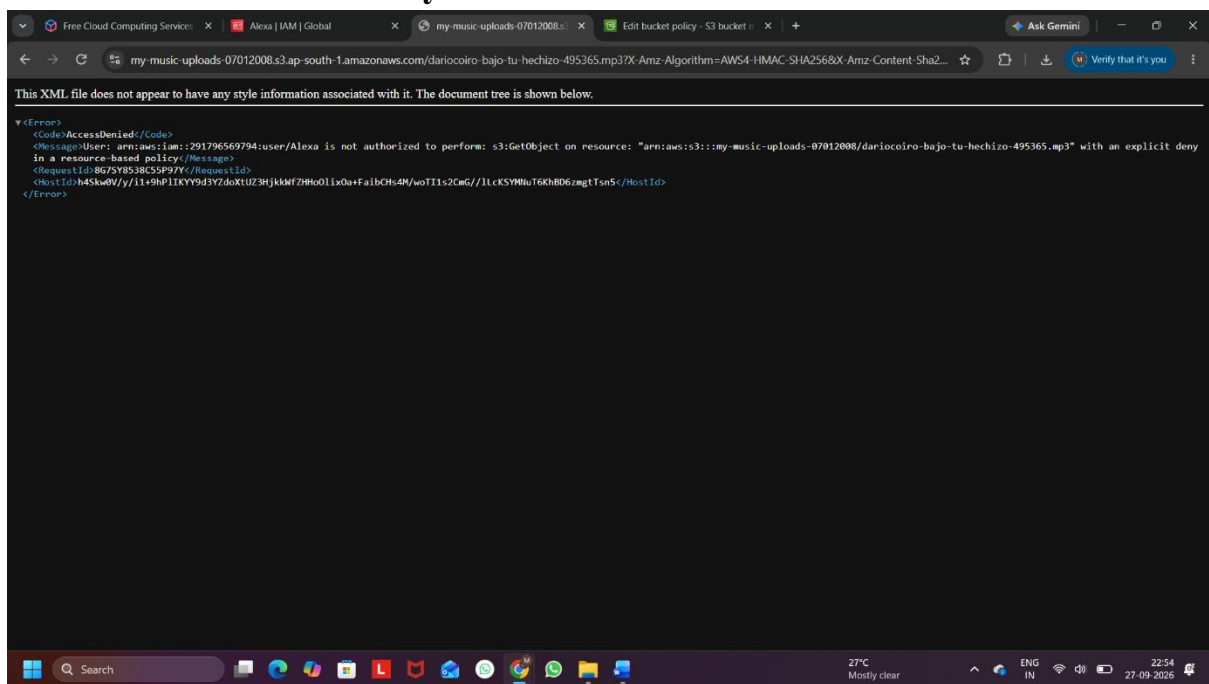
Screenshot 1: Identifying the Public IP Address



Screenshot 2: Applying IP-Based Access Restriction to S3 Bucket



Screenshot 3: Successfully Downloaded MP3 from Authorized IP Address



Screenshot 4: Access Denied from Unauthorized IP Address
Result:

The S3 bucket was successfully configured to allow downloads from the specified IP address and deny access from other IP addresses.